Dohyun (Eric) Kim

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PROFILE

Penultimate Computer Systems Engineering (Hons) student at the University of Auckland with a Mechatronics adjacent focus on embedded systems and digital hardware design. Research assistant in UoA's robotics lab, with hands-on experience in PCB design with Altium, VHDL on FPGA, and AVR based embedded systems.

EDUCATION

University of Auckland – Auckland CBD, New Zealand *Expected Graduation Nov 2027*

- Degree: *Bachelor of Engineering (Honours) in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCES

Research Assistant **Auckland, New Zealand**

University of Auckland *Jul 2026 - Present*

- Develop and test **navigation and control algorithms** for **TurtleBot2** robots in a UoA robotics lab (CARES), including sensor integration and performance evaluation.
- Assist with experiment design, data collection, and analysis **to support ongoing robotics research**.

Robotics Instructor **Auckland, New Zealand**

Creative Imaginary Lab (ciLab) *Apr 2026 - Present*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

PROJECTS

Smart Energy Monitor **Embedded C & PCB**

Embedded Energy Monitoring Systems - Team Project *Oct 2025*

- Designed an embedded energy monitoring system using **ATmega328P microcontrollers** with ADC processing to **measure and display real-time energy usage.**
- Produced a validated PCB schematic using **Altium Designer** to achieve accurate signal conditioning.

Flappy Universe **VHDL & FPGA**

FPGA Game Implementation - Team Project *Jun 2026*

- Implemented a Flappy Bird-style game in VHDL on an **Altera FPGA** with VGA output and sprite rendering to **demonstrate real-time hardware design**.
- Developed a pixel-priority VHDL pipeline in order to **render game scenes at VGA resolution**.

AWARD

3rd Place in ECSE Design Competition 2025 **Team Competition**

"Winnie the Bot" - Interactive companion robot *Sep 2025*

- Built an AI-powered robot using **dual ATmega328P microcontrollers**, an AI camera, and audio peripherals in order to enable **face tracking**, arm movement, and voice dialogue in a compact form factor.
- Contributed to **prototyping the enclosure** using iterative design in order to deliver a functional physical build.

LEADERSHIPS

Academic Team Executive **Auckland, New Zealand**

KEB (Korean Engineering Body) - Founding Member *Jul 2024 - Present*

- Developed and delivered tutorial sessions for **20+ junior engineering students** across various courses.
- Participated in the planning and delivery of academic events **to support the student community**.

Official Competition Staff **NZ Robotics Olympiad 2026**

IEEE R&A (IEEE Robotics and Automation Society) *Jul 2026*

- **Selected as official Competition Staff** for NZRO 2026; ran operations and troubleshoot technical issues.

Full-time Student Volunteer **NZ Robotics Olympiad 2025**

IEEE (Institute of Electrical and Electronics Engineers) *Jul 2025*

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.

SKILLS

Languages & Frameworks: C, VHDL, Python, MATLAB, R, Java, React.js

Tools & Platforms: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, AutoCAD, Git, AWS